

Window Function

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Lab1

The screenshot shows the pgAdmin 4 interface. The left pane displays the database structure, with 'country' selected under 'Tables (17)'. The main pane shows a SQL query using the `DENSE_RANK()` window function to rank films by length within each category. The query is as follows:

```
1 WITH RankedFilm AS (SELECT c.name AS category, f.title AS film_title, f.length AS film_len,
2 DENSE_RANK() OVER (PARTITION BY c.name ORDER BY f.length DESC, f.title) as rn
3 FROM film f
4 JOIN film_category fc on f.film_id = fc.film_id
5 JOIN category c ON c.category_id = fc.category_id
6 )
7 SELECT * FROM RankedFilm Where rn <=3
```

The 'Data Output' pane shows the results of the query, displaying 8 rows of data. The columns are: category, film_title, film_len, and rn. The results are as follows:

category	film_title	film_len	rn
Action	Damn Forrester	185	1
Action	Worst Banger	185	2
Action	Casualties Encino	179	3
Animation	Gangs Pride	185	1
Animation	Pond Seattle	185	2
Animation	Sons Interview	184	3
Children	Fury Murder	178	1
Children	Wrong Behavior	178	2

The status bar at the bottom indicates 'Total rows: 48' and 'Query complete 00:00:00.141'.

Lab2

The screenshot shows the pgAdmin 4 interface. The left pane displays the database structure, with 'country' selected under 'Tables (17)'. The main pane shows a SQL query using the `LAG()` window function to calculate the growth percentage of monthly revenue. The query is as follows:

```
1 WITH MonthlyRevenue AS (
2 SELECT
3 DATE_TRUNC('month', payment_date) AS payment_month,
4 SUM(amount) AS current_revenue
5 FROM payment
6 GROUP BY DATE_TRUNC('month', payment_date)
7 ),
8 RevenueWithLag AS (
9 SELECT
10 payment_month,
11 current_revenue,
12 LAG(current_revenue) OVER (ORDER BY payment_month) AS previous_revenue
13 FROM MonthlyRevenue
14 )
15 SELECT
16 payment_month,
17 current_revenue,
18 previous_revenue,
19 ((current_revenue - previous_revenue) / previous_revenue) * 100 AS growth_percentage
20 FROM RevenueWithLag;
```

pgAdmin 4

File Object Tools Edit View Window Help

Explorer

- 1.3 Sequences
- Tables (17)
 - actor
 - address
 - category
 - city
 - country
 - customer
 - film
 - film_actor
 - film_category
 - inventory
 - language
 - payment
 - rental
 - rental_log
 - secure_order
 - staff
 - store
- Trigger Functions

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Data Output Messages Notifications

Showing rows: 1 to 4 Page No: 1 of 1

	payment_month timestamp without time zone	current_revenue numeric	previous_revenue numeric	growth_percentage numeric
1	2007-02-01 00:00:00	8351.84	[null]	[null]
2	2007-03-01 00:00:00	23886.56	8351.84	186.0035632866530000
3	2007-04-01 00:00:00	28559.46	23886.56	19.56288389789069669300
4	2007-05-01 00:00:00	514.18	28559.46	-98.19961581906660700200

Lab3

pgAdmin 4

File Object Tools Edit View Window Help

Explorer

- 1.3 Sequences
- Tables (17)
 - actor
 - address
 - category
 - city
 - country
 - customer
 - film
 - film_actor
 - film_category
 - inventory
 - language
 - payment
 - rental
 - rental_log
 - secure_order
 - staff
 - store
- Trigger Functions
- Types
- Views
- Subscriptions
- postgres

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Query Query History

```

1 SELECT
2     customer_id,
3     payment_date,
4     amount,
5     SUM(amount) OVER (
6         PARTITION BY customer_id
7         ORDER BY payment_date
8     ) AS running_total_clv
9 FROM payment
10 ORDER BY customer_id, payment_date;

```

Data Output Messages Notifications

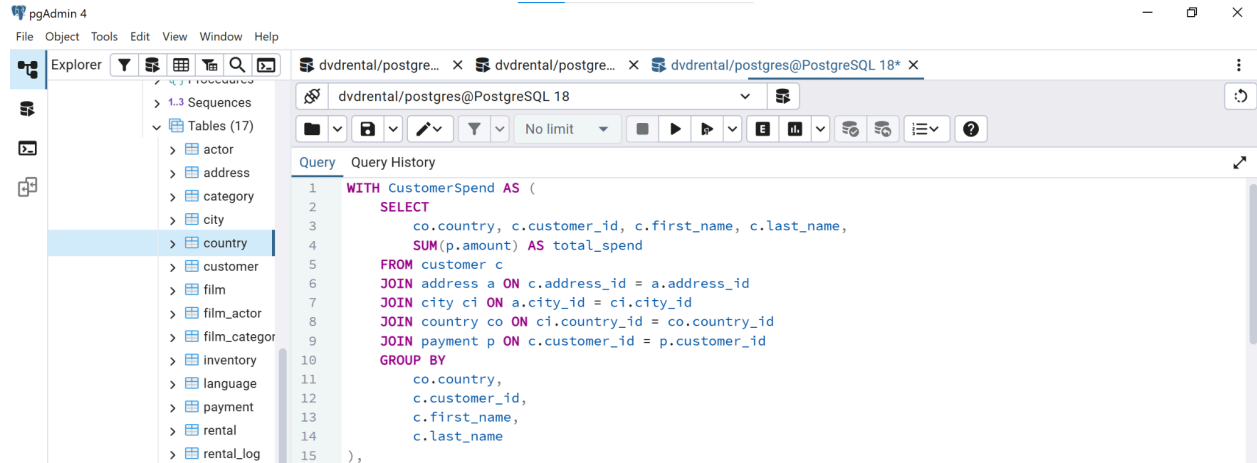
Showing rows: 1 to 1000 Page No: 1 of 15

	customer_id smallint	payment_date timestamp without time zone	amount numeric (5,2)	running_total_clv numeric
1	1	2007-02-14 23:22:38.996577	5.99	5.99
2	1	2007-02-15 16:31:19.996577	0.99	6.98
3	1	2007-02-15 19:37:12.996577	9.99	16.97
4	1	2007-02-16 13:47:23.996577	4.99	21.96
5	1	2007-02-18 07:10:14.996577	4.99	26.95
6	1	2007-02-18 12:02:25.996577	0.99	27.94

Total rows: 14596 Query complete 00:00:00.280 CRLF Ln 10, Col 36

Lab4

Step1:



Step 2,3:

```

RankedCustomers AS (
    SELECT
        country, customer_id, first_name, last_name, total_spend,
        RANK() OVER (PARTITION BY country ORDER BY total_spend DESC) as rank
    FROM CustomerSpend
)
SELECT
    country, customer_id, first_name, last_name, total_spend
FROM RankedCustomers
WHERE rank = 1;

```

